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Research Interests

Climate sensitivity, cloud feedback, radiative forcing

Education

- **Ph.D.** Atmospheric Sciences
University of Washington, 2010
- **M.S.** Atmospheric Sciences
University of Washington, 2007
- **B.S.** Meteorology
Pennsylvania State University, 2004

Professional Experience

- **Research Scientist**
Lawrence Livermore National Laboratory
2013–present
- **Postdoctoral Research Scholar**
Lawrence Livermore National Laboratory
2011–2013
- **Graduate Research Assistant**
University of Washington, Dept. of Atmospheric Sciences
2004–2010

Submitted Publications

Asterisks indicate current or former advisees

- Werapitiya, G., T. Aerenson, D. T. McCoy, G. S. Elsaesser, F. Brient, C. Song, **M. D. Zelinka**, 2026: Origins of the persistent cloud feedback dipole in the hydrological cycle, submitted.
- Myers, T. A*, T. Aerenson, D. T. McCoy, **M. D. Zelinka**, J. M. Nugent, H. Brown, 2026: Cloud-Controlling Factors Reveal Mean-State Controls and Structural Biases in Earth System Model Stratocumulus Physics, *Geophys. Res. Lett.*, submitted.
- Beydoun, H., N. Mahfouz, A. Varble, C. R. Terai, L.-W. Chao*, **M. D. Zelinka**, L. Bertagna, N. D. Keen, P. M. Caldwell, 2026: Thermodynamic Buffering of the Tropical Cloud Fraction Response to Warming in a Global Storm-Resolving Model, *Geophys. Res. Lett.*, submitted.
- Norris, J. S., Rahimi, L., Huang, C., Thackeray, B., Bass, M., **M. D. Zelinka**, and A. Hall, 2026: High vulnerability of Western United States mountain ranges to heavy rainfall events under warming, *Commun. Earth Environ.*, submitted.
- Ceppi, P., A. Bodas-Salcedo, **M. D. Zelinka**, T. Andrews, F. Brient, R. Chadwick, J. M. Gregory, Y.-T. Hwang, S. M. Kang, J. E. Kay, T. Mauritsen, T. Ogura, G. Tselioudis, M. Watanabe, M. J. Webb, and A. A. Wing, 2026: The Cloud Feedback Model Intercomparison Project (CFMIP) contribution to CMIP7, *Geosci. Model Dev.*, submitted.
- Thackeray, C., J. Norris, A. Hall, **M. D. Zelinka**, and S. Po-Chedley, 2026: Record setting heavy precipitation occurrence in 2024, *Sci. Adv.*, submitted.

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111. Van Loon, S., M. Rugenstein, **M. D. Zelinka**, and T. Andrews, 2026: Recent Weakening of the Global Radiative Feedback, *Geophys. Res. Lett.*, 53, doi:10.1029/2026GL122980.
110. Dingley, B., et al. including **M. D. Zelinka**, 2026: CMIP7 Data Request: Atmosphere Priorities and Opportunities, *Geosci. Model Dev.*, 19, 2945–2984, doi:10.5194/gmd-19-2945-2026.
109. **Zelinka, M. D.**, T. A. Myers*, Y. Qin*, L.-W. Chao*, S. A. Klein, S. Po-Chedley, P.-L. Ma, C. Wall, P. Ceppi, A. Gettelman, 2026: Recent cloud trends and extremes reaffirm established bounds on cloud feedback and aerosol-cloud interactions, *Commun. Earth Environ.*, 7, 615, doi:10.1038/s43247-026-03461-8.
108. Tam, R. Y. S., T. A. Myers*, **M. D. Zelinka**, C. Proistosescu, Y.-J. Lin, K. Marvel, 2026: Meteorological Drivers of the Low-Cloud Radiative Feedback Pattern Effect and its Uncertainty, *Atmos. Chem. Phys.*, 26, 4289–4311, doi:10.5194/acp-26-4289-2026.
107. Chao, L.-W.*, **M. D. Zelinka**, C. R. Terai, H. Beydoun, B. R. Hillman, N. D. Keen, P. M. Caldwell, P. A. Bogenschutz, and S. A. Klein, 2026: Examining Cloud Feedback Components in the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM), *J. Climate*, 39, 2563–2578, doi: 10.1175/JCLI-D-25-0656.1.
106. Zhang, Y., et al. including **M. D. Zelinka**, 2026: Understanding Biases in Simulated Cloud Radiative Effects in E3SMv3, *Earth Space Sci.*, doi:10.1029/2025EA004757.
105. Feng, C., et al. including **M. D. Zelinka**, 2026: Interconnection of Aerosol Cloud Interactions and Cloud Feedback through Warm Rain Process, *J. Geophys. Res.*, 131, doi:10.1029/2025JD044854.
104. Ceppi, P., S. Wilson Kemsley, H. Andersen, T. Andrews, R. J. Kramer, P. Nowack, C. J. Wall, and **M. D. Zelinka**, 2026: Emerging low-cloud feedback and adjustment in global satellite observations, *Atmos. Chem. Phys.*, 26, 4153–4171, doi:10.5194/acp-26-4153-2026.
103. Espinosa, Z*, **M. D. Zelinka**, C. M. Bitz, K. C. Armour, 2026: The Strength of Coupling to the Southern Ocean Modulates Tropical Eastern Pacific Variability and Forced Response, *J. Climate*, 39, 1543–1558, doi:10.1175/JCLI-D-25-0226.1.
102. Nugent, J. M., et al. including **M. D. Zelinka**, 2026: Overview of the Nephela Perturbed Parameter Ensemble for aerosol-cloud interactions in E3SMv3, *J. Adv. Model. Earth Syst.*, 18, doi:10.1029/2025MS004989.
101. Fridlind, A. M., et al. including **M. D. Zelinka**, 2025: Toward a Climate OSSE Framework for Satellite Mission Design, *Bull. Amer. Meteor. Soc.*, 107, E431–E445, doi:10.1175/BAMS-D-24-0242.1.
100. Aeronson, T. et al. including **M. D. Zelinka**, 2025: Global models predict clouds at the wrong time of day: does it matter for radiation and climate? *Sci. Adv.*, 11, eady3236, doi:10.1126/sciadv.ady3236.
99. Zehrung, A., A. D. King, Z. Nicholls, **M. D. Zelinka**, and M. Meinshausen, 2025: Standardising the “Gregory method” for calculating equilibrium climate sensitivity, *Geosci. Model Dev.*, 18, 9433–9450, doi:10.5194/gmd-18-9433-2025.
98. Chao, L.-W.*, **M. D. Zelinka**, et al., 2025: Evaluating Mean State Cloud Properties in the Simple Cloud-Resolving E3SM Atmosphere Model (SCREAM), *J. Adv. Model. Earth Syst.*, 17, doi:10.1029/2025MS005133.
97. Hill, P. G., D. L. Finney, and **M. D. Zelinka**, 2025: Cloud feedback uncertainty in the equatorial Pacific across CMIP6 models, *Geophys. Res. Lett.*, 52, doi:10.1029/2025GL117183.
96. Terai, C. R., et al. including **M. D. Zelinka**, 2025: Radiative, hydrologic, and circulation responses to warming in Cess-Potter simulations using the global 3.25-km SCREAM, *J. Adv. Model. Earth Syst.*, 17, doi:10.1029/2024MS004930.
95. Qin, Y*, P.-L. Ma, **M. D. Zelinka**, S. A. Klein, T. Zhang, X. Zheng, V. E. Larson, M. Huang, 2025: Impact of Turbulence Representation on the Relationship Between Cloud Feedback and Aerosol-Cloud Interaction in an E3SMv2 Perturbed Parameter Ensemble, *J. Adv. Model. Earth Syst.*, 17, doi:10.1029/2024MS004756.
94. Mauritsen, T., et al. including **M. D. Zelinka**, 2025: Earth’s energy imbalance more than doubled in recent decades, *AGU adv.*, 6, doi:10.1029/2024AV001636.
93. Zhou, C*, Q. Wang, I. Tan, L. Zhang, **M. D. Zelinka**, M. Wang, and J. Bloch-Johnson, 2025: Sea ice pattern effect on Earth’s energy budget is characterized by hemispheric asymmetry, *Sci. Adv.*, 11, doi:10.1126/sciadv.adr4248.
92. **Zelinka, M. D.**, L.-W. Chao*, T. A. Myers*, Y. Qin*, and S. A. Klein, 2025: Technical note: Recommendations for diagnosing cloud feedbacks and rapid cloud adjustments using cloud radiative kernels, *Atmos. Chem. Phys.*, 25, 1477–1495, doi:10.5194/acp-25-1477-2025.
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- interactions with other climate feedbacks, *Environ. Res.: Climate*, 4, 015001, doi:10.1088/2752-5295/ada84b.
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 87. Espinosa, Z.* and **M. D. Zelinka**, 2024: The Shortwave Cloud-SST Feedback Amplifies Multi-Decadal Pacific Sea Surface Temperature Trends: Implications for Observed Cooling, *Geophys. Res. Lett.*, 51, doi:10.1029/2024GL111039.
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 81. Qin, Y.*, X. Zheng, S. A. Klein, **M. D. Zelinka**, P.-L. Ma, J.-C. Golaz, S. Xie, 2024: Causes of Reduced Climate Sensitivity in E3SM from Version 1 to Version 2, *J. Adv. Model. Earth Syst.*, 16, doi:10.1029/2023MS003875.
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 79. Samset, B., C. Zhou*, J. S. Fuglestad, M. T. Lund, J. Marotzke, and **M. D. Zelinka**, 2023: Steady global surface warming from 1973 to 2022 but increased warming rate after 1990, *Commun. Earth Environ.*, 4, 400, doi:10.1038/s43247-023-01061-4.
 78. Bonan, D. B., N. Feldl, **M. D. Zelinka**, and L. C. Hahn, 2023: Contributions to regional precipitation change and its polar-amplified pattern under warming, *Environ. Res.: Climate*, 2, 035010, doi:10.1088/2752-5295/ace27a.
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- Simulations for Radiative Feedbacks and Forcing from CO₂, *J. Geophys. Res.*, 127, doi:10.1029/2021JD035460.
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45. Terai, C. R., Y. Zhang, S. A. Klein, **M. D. Zelinka**, J. C. Chiu, and Q. Min, 2019: Mechanisms behind the extratropical stratiform low-cloud optical depth response to temperature in ARM site observations, *J. Geophys. Res.*, 124, doi:10.1029/2018JD029359.
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Book Chapters

6. Chao, L. -W*, A. E. Dessler, and **M. D. Zelinka**, 2026: Climate Feedbacks. In: Robinson, W.A. (Ed.), *Encyclopedia of Atmospheric Sciences*, vol. 5. Elsevier, Academic Press, pp. 117–128, doi:10.1016/B978-0-323-96026-7.00038-2.
5. McCoy, D. T., M. E. Frazer, J. Muelmenstaedt, I. Tan, C. R. Terai, and **M. D. Zelinka**, 2024: Extratropical Cloud Feedbacks, in *Clouds and their Climatic Impacts: Radiation, Circulation, and Precipitation*, S. C. Sullivan (Ed) and C. Hoose (Ed), American Geophysical Union.
4. Contributing author to Forster et al. 2021: The Earth's Energy Budget, Climate Feedbacks, and Climate Sensitivity, in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Masson-Delmotte et al. (eds.). Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, pp. 923-1054, doi:10.1017/9781009157896.009.
3. McCoy, D.T., D. L. Hartmann, and **M. D. Zelinka**, 2017: Mixed-Phase Cloud Feedbacks, in *Mixed-phase Clouds: Observations and Modeling*, Andronache, C. (Ed.), Elsevier.
2. Tan, I., T. Storelvmo, and **M. D. Zelinka**, 2017: The climatic impact of thermodynamic phase partitioning in mixed-phase clouds, in *Mixed-phase Clouds: Observations and Modeling*, Andronache, C. (Ed.), Elsevier.
1. Dessler, A.E. and **M. D. Zelinka**, 2015: Climate Feedbacks, in *Encyclopedia of Atmospheric Sciences, 2nd edition*, Vol 2, pp. 18-25, G. R. North (editor-in-chief), J. Pyle and F. Zhang (editors).

Recent Honors & Awards

- Richard P. and Linda S. Turco Lectureship, Department of Climate, Meteorology, & Atmospheric Sciences, University of Illinois Urbana–Champaign, 2024-2025.
- LLNL Physical and Life Sciences Directorate Awards for Excellence in Publications for Terai et al. (2025), Lee et al. (2024), Sherwood et al. (2020), Zhou et al. (2016), and Norris et al. (2016)
- Zelinka et al. (2020) named "Editor's Choice Paper" by Editor-in-Chief of *Geophysical Research Letters*, 2024
- American Meteorological Society Henry G. Houghton Award, 2022
- Eos Research Spotlights for Zelinka et al. (2022), Zelinka et al. (2020), Zhou et al. (2017), Zelinka et al. (2016), and McCoy et al. (2016)
- Editors' Citation for Excellence in Refereeing - *Geophysical Research Letters*, 2021
- LLNL Deputy Director's Science & Technology Excellence in Publication Awards for Zelinka et al. (2020), Sherwood et al. (2020), and Santer et al. (2015)
- Sherwood et al. (2020) named runner-up for *Science Magazine's* 2020 Breakthrough of the Year
- *Nature Climate Change* Research Highlight for Dong et al. (2020)
- US CLIVAR Research Highlight for Zelinka et al. (2020)
- LLNL Outstanding Mentor Award, 2018
- LLNL Early and Mid-Career Recognition Program Award, 2018
- *Diablo Magazine* 40 Under 40, 2017

Professional Service and Leadership

- Co-chair, 2027 Gordon Research Conference on Radiation and Climate
- Vice Chair, 2025 Gordon Research Conference on Radiation and Climate
- Editor, *Journal of Climate*, Dec 2024–present
- Steering Group, CMIP7 Data Request (Atmosphere Theme), Jul 2024–present
- External Peer Reviewer, NASA Langley Research Center Science Directorate, Nov 2023
- Programme Advisory Group, Uncertainty in Climate Sensitivity due to Clouds (CloudSense) Research Programme, Sep 2023–present
- CFMIP Scientific Steering Committee, Jul 2023–present
- Discussion Leader, 2023 Gordon Research Conference on Radiation & Climate
- AGU Global Environmental Change Fellows Committee, 2022–2024
- Contributing Author, IPCC 6th Assessment Report, 2019–2021

- LLNL Physical and Life Sciences Postdoc Committee, 2017–2020
- Section Editor, *Current Climate Change Reports* Topical Collection on Climate Feedbacks, 2014–2017
- Contributor, climatefeedback.org, 2017–present
- Chair, 2011 Gordon Research Seminar on Radiation and Climate
- Proposal reviewer for DOE, European Research Council, NASA, and NSF
- Editor, *Proceedings of the National Academy of Sciences*
- Reviewer for:

<i>Atmosphere</i>	<i>Atmos. Ocean</i>	<i>Atmos. Chem. Phys.</i>	<i>Atmos. Meas. Tech.</i>
<i>Atmos. Sci. Lett.</i>	<i>B Am Meteorol Soc</i>	<i>Clim. Dynam.</i>	<i>Climatic Change</i>
<i>Earth's Future</i>	<i>Earth Syst. Dyn.</i>	<i>Earth Syst. Sci. Data</i>	<i>Environ. Res. Lett.</i>
<i>Geophys. Res. Lett.</i>	<i>Geosci. Model Dev.</i>	<i>Int. J. Appl. Earth Obs. Geoinf.</i>	<i>J. Adv. Model. Earth Syst.</i>
<i>J. Appl. Meteorol. Clim.</i>	<i>J. Atmos. Oceanic Technol.</i>	<i>J. Atmos. Sci.</i>	<i>J. Climate</i>
<i>J. Geophys. Res</i>	<i>J. Meteorol. Soc. Jpn.</i>	<i>Nature</i>	<i>Nat. Clim. Change</i>
<i>Nat. Commun.</i>	<i>Nat. Geosci.</i>	<i>npj Clim. Atmos. Sci.</i>	<i>P. Natl. Acad. Sci.</i>
<i>Sci. Rep.</i>	<i>Sci. Adv.</i>	<i>Surv. Geophys.</i>	

Advising

Postdoctoral Researchers

- Li-Wei Chao (Postdoc, Tufts University)
- Yi Qin (Research Scientist, Laoshan Laboratory)
- Timothy Myers (Research Scientist, University of Wyoming)
- Chen Zhou (Associate Professor, Nanjing University)

Graduate Students

- Zac Espinosa (Google DeepMind)

Undergraduate Students

- Russell Hunter (Duke University / Second Lieutenant, US Space Force)
- Thea Moellerstedt (Graduate Student, UCSD)
- Scott Feldman (Meteorologist, Verisk Weather Solutions)

Recent Invited Presentations

- ESA-JAXA EarthCARE Workshop (Keynote), 8 Jun 2026
- GEWEX Earth's Energy Imbalance Assessment Workshop, 4 Jun 2026
- Libera Science Team Meeting, 19 May 2026
- UC San Diego, Scripps Institution of Oceanography, 27 Jan 2026
- McGill University, Department of Atmospheric and Oceanic Sciences, 21 Oct 2025
- Duke University, Earth & Climate Sciences Division, 3 Oct 2025
- Stanford University, Department of Earth System Science, 2 Apr 2025
- Univ. of Illinois Urbana–Champaign, Dept. of Climate, Meteorology, & Atmospheric Sciences, 22 Oct 2024
- CERES Science Team Meeting, 2 Oct 2024
- LLNL Senior Staff Meeting, 1 Jul 2024
- NASA Ames Research Center, Earth Science Division, 20 Jun 2024
- Yale University, School of the Environment, 25 Apr 2024
- AGU Fall Meeting: The Flows of Energy Through the Climate System Session, 12 Dec 2023
- AGU Fall Meeting: Climate Sensitivity and Feedbacks Session, 11 Dec 2023
- University of Cambridge Centre for Atmospheric Science, Department of Chemistry, 24 Oct 2023

- Yale University School of the Environment, 27 Apr 2023
- AGU Fall Meeting: Atmospheric Physics, Radiation, Clouds, and Aerosols Session, 15 Dec 2022
- Yale University, School of the Environment, 21 Apr 2022
- Aerosol and Cloud, Convection and Precipitation Webinar Series, 19 Apr 2021
- University of Maryland Baltimore County Department of Physics Colloquium, 24 Feb 2021
- University of Toronto Physics Colloquium, 28 Jan 2021

Outreach

- LLNL Ambassador Lecturer, Aug 2024 - present
- LLNL Technical Lightning Talk: Community Cohorts program for new hires, 24 Aug 2023.
- LLNL - Las Positas College Science and Engineering Seminar Series, 20 Apr 2023.
- Science on Saturday, The Future in Focus: Predicting Climate Change through Observations, Modeling, and Artificial Intelligence, 4 Mar 2023.
- Castro Valley Rotary Club, 13 Sep 2022.
- Science on Saturday, The Future in Focus: Predicting Climate Change through Observations, Modeling, and Artificial Intelligence, 26 Feb 2022.
- San Joaquin County Office of Education Climate Change Summit, 26 Sep 2020
- Panel Discussant, Wild and Scenic Film Festival, Bankhead Theater, Livermore, CA, Jan 2020
- Univ. of Washington Dept. of Atmospheric Sciences and Program on Climate Change Outreach Teams, Sep 2004 – Dec 2010

Last updated: July 31, 2026